

Concept Summary: The Memory Cliff — Associative Memory and Fast Weights

Track: DataForge 2026 — Pathway Track

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Core Topic: Associative Memory and Fast Weights in Recurrent and Linear-Attention Architectures

1. The Falsifiable Claim & Scientific Motivation

Core Claim: “A *fixed-size additive associative-memory state* can process an arbitrarily long stream of facts without allocating a new slot per fact, but its exact-recall accuracy degrades once the number of stored facts exceeds the state’s effective capacity, due to key-collision interference.”

Standard softmax attention maintains an explicit Key-Value (KV) cache proportional to sequence length T , creating an $\mathcal{O}(T)$ memory footprint that limits long contexts.

Recurrent sub-quadratic architectures—such as linear attention, DeltaNet, and the Dragon Hatchling (BDH)—compress history into a **constant-size state matrix** $W_t \in \mathbb{R}^{d \times d}$. With $\mathcal{O}(1)$ recurrent updates, they ingest unbounded token streams without per-token cache allocations.

However, linear algebra imposes **the Memory Cliff**: a fixed matrix with linear readout ($y = Wq$) stores at most d mutually orthogonal keys. When stored facts N exceed dimension d , keys become linearly dependent in $\mathbb{R}^d$, producing cross-talk interference that degrades recall toward zero. Our interactive substrate tests, visualizes, and diagnoses this boundary.

2. Experimental Architecture: Three Compared Models

Our substrate evaluates three architectures on a synthetic associative recall task ($d = 32$, $N \in [1, 96]$):

1. Full Attention Baseline (KV Cache):

- *Mechanism:* Explicit (k_i, v_i) caching with dot-product readout: $y = \text{Softmax}(qK^\top)V$.
- *Behavior:* Maintains 100% recall, incurring $\mathcal{O}(N)$ memory costs ($2 \cdot N \cdot d$ floats).

2. Fixed Memory — Additive Fast-Weights (Primary BDH Analogue):

- *Mechanism:* Outer-product write rule: $W_t = W_{t-1} + v_t k_t^\top$, with linear readout $y = W_t q$.
- *Behavior:* Constant 1,024-float footprint. High fidelity ($> 95\%$) for $N \leq 32$; collapses for $N > 32$.
- *Labeling:* Mechanistic analogue of BDH’s Hebbian update, not an official BDH model.

3. DeltaNet Contrast (Corrective Delta Rule):

- *Mechanism:* Subtractive delta update: $W_t = W_{t-1} + \beta(v_t - W_{t-1}k_t)k_t^\top$.
 - *Behavior:* Constant 1,024-float footprint; subtracts prediction errors to dampen collision interference.
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3. The BDH Connection: Biological Synaptic Plasticity & Fast Weights

Our project connects fast weights to neurobiological foundations in Dragon Hatchling (BDH) research:

- **Hebbian Plasticity Update:** In *The Dragon Hatchling* (Kosowski et al., 2025, arXiv:2509.26507), attention is reformulated as biological synaptic plasticity: $W_t = W_{t-1} + \eta \cdot (y_{\text{post}} \otimes x_{\text{pre}})$. Synapses strengthen from correlated activations without negative feedback. Our Fixed Memory model isolates this additive rule.
 - **Recurrent Demonstration Ingestion:** In *BDH-CQ* (Engdahl et al., 2026, arXiv:2608.09888), in-context learning sequentially updates a recurrent state across demonstration exemplars without a KV cache. Our project models this via an idealized additive fast-weight update.
 - **Architectural Distinction:** DeltaNet cannot serve as a BDH stand-in: DeltaNet relies on subtractive error correction, whereas BDH is grounded in purely additive Hebbian reinforcement.
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4. Classification of Evidence Types

Per evaluation standards, we explicitly classify each major result’s evidence type:

- **Arora et al. (2024, arXiv:2402.18668, “Based”)**: *Academic benchmark* establishing state-size vs. recall tradeoffs across synthetic tasks and models up to 1.3B parameters.
 - **Yang et al. (2024, arXiv:2412.06464, Gated DeltaNet)**: *Academic benchmark* showing subtractive delta-rule updates outperform uncorrected linear states on recall.
 - **Pandey & Singh (2026, arXiv:2605.11196, Variational Linear Attention)**: *Academic benchmark* introducing regularized least-squares updates to sustain recall up to head capacity ($n_{\text{pairs}} \leq d_h$).
 - **Kosowski et al. (2025, arXiv:2509.26507, BDH)**: *Developer-reported study* by Pathway (not an independent reproduction) detailing Hebbian synaptic dynamics and language scaling.
 - **Engdahl et al. (2026, arXiv:2608.09888, BDH-CQ)**: *Developer-reported evaluation* by Pathway demonstrating recurrent latent reasoning on ARC-AGI-1 (\$0.0007/task).
 - **DataForge Interactive Dashboard** ($d = 32, N \in [1, 96]$): *Live synthetic measurement* executing transparent forward passes on an isolated mathematical analogue, not official BDH checkpoints.
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5. Interactive Substrate & Verification

All components execute live PyTorch or deterministic simulations:

- **Live Per-Fact Grid**: Evaluates recall in real time (green/red cosine similarity).
 - **Capacity Slider**: Sweeps the $d = 32$ limit to observe interference onset.
 - **State Microscope**: Exposes 32×32 tensor activations.
 - **Demonstration Surgery**: Ablates/corrupts facts and updates the state in real time.
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6. Known Limitations

- **Hand-Crafted Analogue, Not an Official BDH Run**: The Fixed Memory model is an isolated mathematical analogue of the additive outer-product Hebbian write rule ($W_t = W_{t-1} + v_t k_t^\top$) inspired by *The Dragon Hatchling* (arXiv:2509.26507). It does not execute Pathway’s official BDH or BDH-CQ pretrained checkpoints, nor does it incorporate BDH’s multi-layer spiking integrate-and-fire thresholding or proprietary recurrent latent workspace dynamics. Furthermore, no matched-scale baseline against trained BDH weights exists, and the $d = 32$ threshold is an educational toy scale chosen for live in-browser execution rather than production LLM dimensions ($d = 2, 048$ to $8, 192$).
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7. Primary Literature References

1. Yang, S., Kautz, J., & Hatamizadeh, A. (2024). *Gated Delta Networks: Improving Mamba2 with Delta Rule*. arXiv:2412.06464.
2. Arora, S., et al. (2024). *Simple linear attention language models balance the recall-throughput tradeoff*. arXiv:2402.18668.
3. Pandey, V., & Singh, G. (2026). *Variational Linear Attention: Stable Associative Memory for Long-Context Transformers*. arXiv:2605.11196.
4. Kosowski, A., et al. (2025). *The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain*. arXiv:2509.26507.
5. Engdahl, B., et al. (2026). *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*. arXiv:2608.09888.